Mialimanana Andrianina Bien-Aimé Razafindrakoto | CV

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Mathematician and machine learning researcher with an MSc in AI for Science from AIMS South Africa. Working in the intersection of pure mathematics, machine learning methods, particularly kernel methods, non-parametric hypothesis testing, and linking statistical learning to functional analysis. Interested in pursuing PhD in Machine Learning and research that connects mathematical theory with emerging methods in machine learning and uncertainty

Education

Stellenbosch University – African Institute for Mathematical Sciences **South Africa**
MSc in AI for Science *September 2025 – July 2026*
Syllabus: Evolution of Deep Learning, Mathematical Foundation of Machine Learning, Machine Learning at Scale, Tensor Decomposition, Reinforcement Learning

University of Antananarivo, Ankatso **Madagascar**
Double MSc Program *January 2023 – September 2025*
- Department of Pure Mathematics
- Department of Mathematics, Algorithmics, Finance and Informatics

Syllabus: Functional Analysis, Commutative Algebra, Number Theory, Machine Learning, Cryptography & Coding Theory, Complex Analysis, Differential Geometry

University of Antananarivo, Ankatso **Madagascar**
Bachelor's Degree in Pure Mathematics *2022 – 2023*
Syllabus: Abstract Algebra, Measure Theory, Topology, Numerical Analysis, Differential Calculus

Research Experience & Projects

Quasi-Monte Carlo Methods for Kernel Quantile Embeddings
Master's Thesis – AIMS Research Project, *Supervisor: Dr. Krikamol Muandet* *Full-time Research Phase*

- Addressed the computational limitations of kernel-based hypothesis testing by integrating Quasi-Monte Carlo (QMC) sampling, Sobol' and Halton sequences, into Kernel Quantile Embedding methods which use standard Monte-Carlo Methods.
- Developed and benchmarked QMC-based testing variants against established non-parametric tests, including Maximum Mean Discrepancy (MMD) and Kernel Quantile Embedding (KQE).
- Improved statistical testing power by up to 20% over baseline methods, depending on datasets and sample-size, while maintaining controlled Type I error rates.

Jury Decision: Pass with Distinction

Research Interest

- Machine Learning/Deep Learning, Kernel Methods
- Non-parametric Hypothesis Testing, Data Science
- Mathematics: Functional Analysis, Probability Theory

Awards & Honors

Google DeepMind Scholarship, MSc AI for Science **South Africa**
MSc awarded cum laude *July 2026*

Summer Schools & Workshops

- **IndabaX - Cyberfest - FCOI**, Madagascar, 29th October - 1st November

Poster Presentation on *Quasi-Monte Carlo Methods for Kernel Quantile Embeddings*
- **GCI World 2026**, Japan, In progress
- **Deep Learning Indaba**, Nigeria, August 2026
- **The Theory of Quantum Learning Algorithms**, South Africa, October 2025

Technical Skills

- Python (NumPy, JAX, PyTorch, Keras, Tensorflow, Pandas, Scipy, XGBoost), R, C
- Tools: LaTeX, Git/GitHub

Languages

- Malagasy: Native Language
- English: Fluent
- French: Professional proficiency, DELF B2